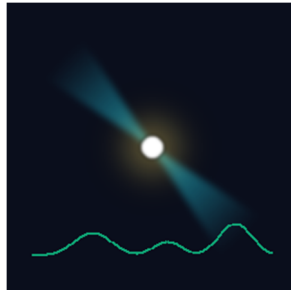


Earth-Venus-Earth Modem

Design Description and Interface Control Document



Rev A - DRAFT

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DSES

September 2026

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Document	DSES EVE Modem Design and ICD
Revision	Rev A — DRAFT for team review
Date	2026-09-10
Prepared by	Rick Hambly, K0GD, Deep Space Exploration Society
Waveform design	Pete Wyckoff, KA3WCA, Open Research Institute (“Venus Bounce Transmitter Spiral #2”)
Status	Design baseline; parameters marked TBC await confirmation from ORI. Draft 2 adds the band plan of 2026-09-10 (23 cm at 1299.5 MHz for Venus 2026; 13 cm later; EME on both bands without amplifiers)

This document describes the DSES implementation of the ORI Earth-Venus-Earth (EVE) waveform for the Venus inferior conjunction of 24 October 2026, and it defines every interface the implementation touches: the signal on the air, the station hardware, the timing and frequency references, the files exchanged with other stations, and the boundary with the DSES Radio Astronomy Workbench. Sections 1 through 5 are the design description. Sections 6 through 9 are the interface control document (ICD). Section 10 is the decision and open-issue register. Anything in this document that another team member depends on is in the ICD sections, so that a change there is a change everyone sees.

The ORI reference material is the EVE repository on GitHub (OpenResearchInstitute/EVE): the MATLAB simulation in `signal_design/` (May–June 2026), the Python transmit generator in `signal_design/Python_Implementation/` (August 2026, the current implementation per ORI), the ORI link budget notebook, and the CAMRAS validation white paper. All ORI code is GPL-3.0 and is credited to Pete Wyckoff and ORI wherever it is used.

1. Purpose and summary

The DSES 60-foot dish at Haswell will attempt to bounce a digital message off Venus near inferior conjunction and decode the echo. ORI designed the waveform and validated it in simulation, and ORI has a transmit-side generator that writes the waveform to a SigMF file for playback through a USRP B210 in GNU Radio. ORI has no receiver. DSES needs a complete, runnable modem: a transmitter that fits the DSES station’s duty cycle and keying, a receiver that decodes echoes, and the synchronization, Doppler, and scheduling machinery that the reference design assumes away.

The design decisions in this revision:

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- **The modem streams.** The transmitter synthesizes the waveform on the fly, driven by a schedule, rather than playing a pre-generated file. SigMF export and import are retained for interoperability with ORI's files, for bench tests, and for sharing with other stations, but they are not how the station transmits (section 3).
- **The 2026 Venus attempt is a 23 cm operation at 1299.5 MHz.** The 13 cm package will not be ready this year, so DSES transmits with the 23 cm package retuned from 1296 to 1299.5 MHz, the frequency Dwingeloo, Stockert, and Effelsberg use. The modem is frequency-agile (1296, 1299.5, 2304, 2400 MHz, and others); every frequency-dependent quantity is computed per schedule (section 2.4).
- **Monostatic operation is the baseline,** with reception by a European station as the upside case at 1299.5 MHz. At 23 cm the monostatic link is 2 to 4 dB below the design point; repeat-and-combine over a session recovers that, and a 25 m or 100 m receiver adds 3 or 15 dB (section 2.2).
- **EME testing needs no power amplifier.** The B210's own output into the dish gives a C/N0 of +12 dB-Hz at 1296 MHz and +16 dB-Hz at 2304 MHz off the Moon, so both bands can be tested at and below the design point before either amplifier exists (section 4.6).
- **The 30-minute message is sent in chunks** of up to the Earth-Venus round-trip time (272 s on 24 October), with the station listening for the echo of each chunk while the amplifier cools. The chunk length is a parameter; 4 minutes is the working value, giving one message in about 64 minutes at 47 percent transmit duty (section 4).
- **The ORI Python generator's conventions are the air-interface specification** (section 6). They differ from the earlier MATLAB simulation in bin width, frame count, bit order, and comb placement; ORI confirmed on 9 September that the MATLAB was early simulation and the Python is the current implementation. Two numerical inconsistencies in the Python are resolved here by definition and flagged TBC.
- **The modem is a separate project from the Workbench.** It reuses the Workbench's proven B210 code by factoring that code into an importable module (section 5 and section 9).
- **Doppler is handled on both ends:** ephemeris pre-compensation on transmit so the echo arrives at the nominal frequency, and residual tracking on receive. On conjunction day the two-way Doppler rate at Haswell reaches 0.48 Hz/s, which is 14 tone spacings across one symbol if left uncorrected (section 4.4).
- **Timing comes from GPS,** not a maser. A non-coherent receiver only needs to know which 0.35-second frame belongs to which symbol; GPS time on the B210 and a shared UTC epoch provide that to a millisecond (section 4.3).
- **Validation runs simulation first, then B210 loopback, then a low-power Earth-Moon-Earth test, then Venus** (section 5.4).

2. Mission context

2.1 Geometry on conjunction day

Computed for Haswell (38.3808° N, 103.1561° W, 1311 m) with astropy's built-in ephemeris; the flight software will use a JPL Horizons table for the operational numbers.

Quantity	2026-10-24	Notes
Earth-Venus distance	0.273 AU (40.8 million km)	Minimum of the apparition
Round-trip light time	272 s (4.5 min)	Sets the maximum monostatic chunk length
Venus above 20° at Haswell	15:30 – 21:15 UTC (5.75 h)	Rises 13:25, sets 23:30 UTC
Maximum elevation	34° at 18:30 UTC	
Angular separation from the Sun	about 6°	Solar noise in the sidelobes raises Tsys; quantify in the link budget
Two-way Doppler shift	1299.5 MHz: +5.3 → −2.5 kHz; 2304 MHz: +9.5 → −4.5 kHz; zero at 19:50 UTC	Scales with frequency
Two-way Doppler rate	1299.5 MHz: up to −0.27 Hz/s; 2304 MHz: up to −0.48 Hz/s (18:30 UTC)	45 Hz (8 tone spacings) or 79 Hz (14) across one 165 s symbol

Venus stays within 10 percent of minimum distance from 7 October to 12 November, so the attempt is a multi-week window, not one day. The round-trip time and Doppler numbers above change slowly over that window and the schedule generator recomputes them per session.

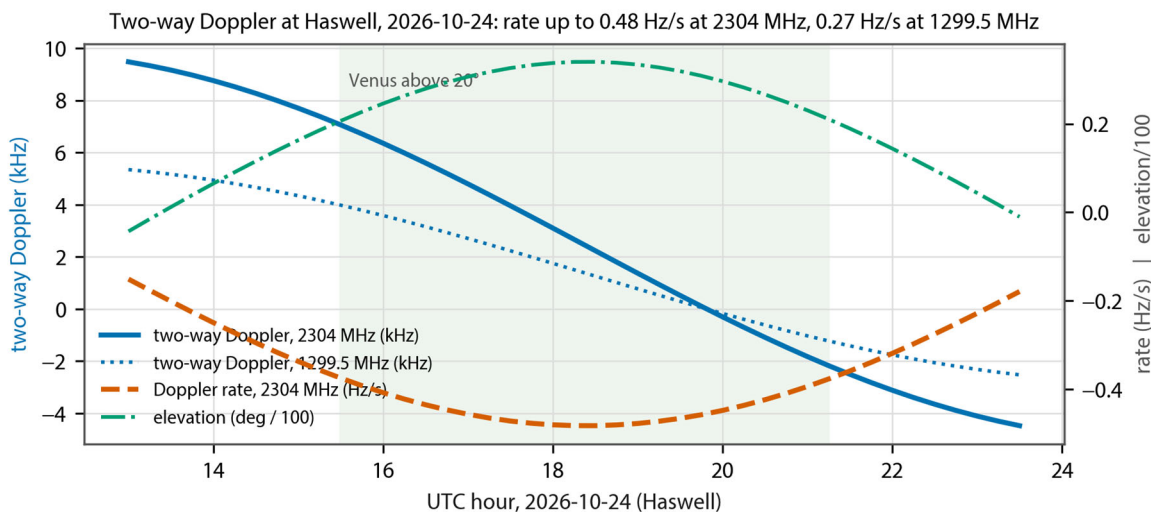


Figure 1 — Two-way Doppler shift at 2304 and 1299.5 MHz, Doppler rate at 2304 MHz, and Venus elevation at Haswell on 24 October 2026. The shaded band is the working window above 20 degrees; at 1299.5 MHz the rate scales by 0.56.

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2.2 Link budget

The ORI link budget notebook's DSES case (18.29 m dish, 69 percent efficiency, 1500 W, 2304 MHz, 0.4 dB LNA, 30° elevation, minimum distance) gives a monostatic C/N0 of +1.7 dB-Hz. Pete Wyckoff sized the waveform for C/N0 = 0 dB-Hz. The 2026 attempt is at 1299.5 MHz with the 23 cm package, so the table scales the ORI case to the bands DSES can actually use, holding the dish and the radar cross-section fixed (the two-way path scales as the wavelength squared, the two antenna gains as its inverse fourth power, net −5 dB at 23 cm):

Case	C/N0 (dB-Hz)	Basis
DSES monostatic, 2304 MHz, 1500 W, Tsys 76 K	+1.7	ORI notebook, DSES dataclass
DSES monostatic, 2400 MHz, 1500 W, Tsys 76 K	+2.0	scaled
DSES monostatic, 1299.5 MHz, 1500 W, Tsys 55 K	−1.8	scaled; albedo 0.152 (CAMRAS)
DSES monostatic, 1299.5 MHz, 1000 W, Tsys 55 K	−3.6	scaled
Cross-check: Dwingeloo's measured +0.65 (25 m, 1 kW, 1299.5 MHz) scaled to an 18.3 m dish	−4.8	gain to the fourth power
DSES transmits at 1299.5 MHz, Dwingeloo or Stockert receives (25 m)	monostatic +2.7	receive gain only
DSES transmits, Effelsberg receives (100 m, Tsys 25 K)	monostatic +15 to +18	ORI proposal: +17.7 dB-Hz at 2304 MHz
Repeat-and-combine, N passes of the message	+10 log10 N	five passes = +7 dB

Three conclusions. First, at 23 cm a single monostatic pass sits 2 to 4 dB below the design point before solar noise, so a single-pass decode is not the plan; five passes in the 5.75-hour window (+7 dB) bring the session to +3 to +5 dB-Hz equivalent, which closes. Second, any European station that receives with the DSES schedule file turns the link positive on its own: Dwingeloo or Stockert by about 3 dB, Effelsberg by 15 dB or more. Third, every decibel of the transmit chain matters at 23 cm: the actual amplifier power, the feed match, and the receiver Tsys are the open items that set where in that table DSES lands (O2). ORI's analytic check of the waveform gives a frame error rate of 2 percent at 0 dB-Hz, 38 percent at −1 dB-Hz, and 60 percent at −1.33 dB-Hz (AWGN); Pete's MATLAB channel adds Rayleigh fading with a 1.05 dB correction, and the DSES receiver validation (section 5.4) includes that so the margin is stated honestly.

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The 13 cm case, when that package exists, is the ORI +1.7 dB-Hz at 2304 MHz or +2.0 at 2400 MHz, with the same repeat-and-combine arithmetic on top.

2.3 Stations and frequencies

Station	Role in October 2026	Frequency
DSES Haswell, 18.3 m	Transmit and receive (this document)	1299.5 MHz in 2026 with the 23 cm package (1296 MHz native; retune TBC, O2). 13 cm package (2304 or 2400 MHz) not before the next apparition
Dwingeloo (CAMRAS), 25 m, 1000 W	Transmit and receive, own campaign	1299.5 MHz only (confirmed by ORI 2026-09-09)
Effelsberg, 100 m	Receive only, opportunistic, secondary to a baseline survey	Follows Dwingeloo; proposal alternates 1299.5 and 2304 MHz
Stockert, 25 m	Receive, own campaign	1299.5 MHz

Because DSES will be on 1299.5 MHz, the European stations can in principle receive DSES during the mutual window, roughly 13:30 to 16:30 UTC at low elevation on both ends (estimate, to be computed exactly when a session is arranged); the schedule file of section 8.1 is what makes that possible. Monostatic remains the baseline because it is the only mode DSES controls end to end.

2.4 Band plan and frequency dependence

The B210 tunes 70 MHz to 6 GHz, and the modem carries the dial frequency in the schedule, so the same software serves every band. What changes with frequency is computed per schedule from the ephemeris; what does not change is the waveform itself, which ORI defines with the same parameters on every band (their own smoke-test file is at 1296 MHz).

Band	Use	Two-way Doppler, Venus at conjunction	Peak rate, per symbol	Doppler spread (ORI: 1.5 Hz at 1299.5, scaled)
1296 MHz	EME tests, 23 cm package native	n/a (Moon: about ± 2 kHz)	Moon: small	libration, variable
1299.5 MHz	Venus 2026 ; Dwingeloo, Stockert, Effelsberg band	+5.3 to -2.5 kHz	0.27 Hz/s, 45 Hz (8 spacings)	1.5 Hz
2304 MHz	13 cm package, next apparition	+9.5 to -4.5 kHz	0.48 Hz/s, 79 Hz (14 spacings)	2.66 Hz
2400 MHz	13 cm alternative	+9.9 to -4.7 kHz	0.50 Hz/s, 83 Hz (14 spacings)	2.77 Hz

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One consequence for the waveform: ORI's 2.87 Hz bin is matched to the spread at 2304 MHz. At 1299.5 MHz the spread is 1.5 Hz, so a 2.87 Hz bin is wider than the channel needs and costs about 2.8 dB relative to a bin matched to 1.5 Hz. A matched 23 cm variant would be a different waveform (tone spacing 3 Hz, symbol 315 s for the same frame count) that the European receivers would have to match; the recommendation is to keep ORI's parameters on every band for interoperability and to book the 2.8 dB against repeat-and-combine, unless ORI decides otherwise (O11). The RF package, feed, and amplifier per band are station items (section 7).

3. Architecture

3.1 The question: pre-generate or stream

ORI's implementation generates the complete waveform as a SigMF file and plays it through a B210 with GNU Radio. For a 30-minute message at 250 kS/s that is a 3.6 GB file, valid for one message and one Doppler profile. DSES will not operate that way, for four reasons:

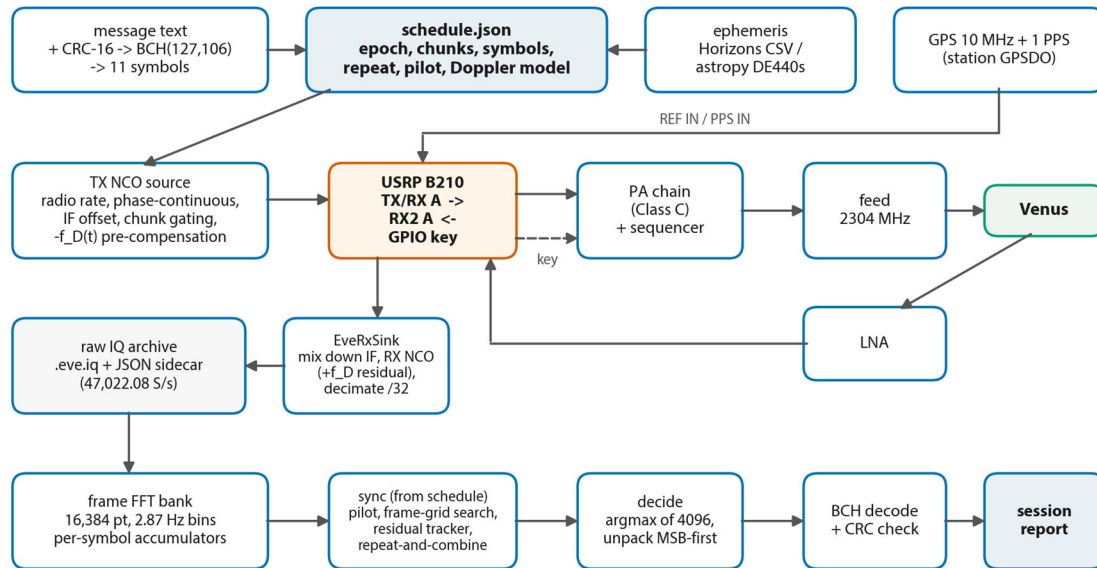
1. **The station transmits in chunks with gaps** (section 4). A pre-generated file has no gaps; cutting it at run time is the same work as generating it at run time.
2. **Doppler pre-compensation changes continuously** and depends on the date, time, and target. A file generated for one session is wrong for the next.
3. **The waveform is trivial to synthesize**: one complex exponential at a time. At the radio's 1.5 MS/s that is a few million complex multiplies per second, well inside a Python NumPy budget, with no resampling because the tone is generated directly at the radio rate.
4. **Nothing is lost**. The transmit log records the schedule and the symbols, which together are a complete description of what went out; the IQ can be regenerated exactly.

The decision is therefore **stream**. The SigMF path is kept as an export (write ORI-format files for other stations and for the bench) and as an import (play an ORI file through the same transmit chain for cross-checks).

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3.2 System block diagram



One B210, one schedule, one Doppler model shared by transmit and receive. The schedule file is the contract with any partner station.

Figure 4 — System block diagram. Transmit and receive chains on one B210, both driven by the schedule file and the same Doppler model.

The transmitter and receiver share one B210, one schedule, and one Doppler model. The schedule is the contract between them and, in bistatic operation, between DSES and the partner station.

3.3 Modes

Mode	Transmit	Receive	Schedule
Monostatic (baseline)	DSES, chunked	DSES, own echo, between chunks	Chunk length $\leq$ round trip; receive window = chunk + round trip
Bistatic transmit	DSES, continuous 30- min message, repeated	Partner station	Continuous, repeat count N; partner receives the schedule file
Bistatic receive	Partner	DSES	DSES receives the partner's schedule file
EME test	DSES, chunks of ≤ 2.5 s (Moon round trip)	DSES	Same machinery, target=moon, shorter symbol
Bench	B210 internal leakage, no PA	B210	Loopback, noise added offline

4. Operating concept

4.1 The message and its timing

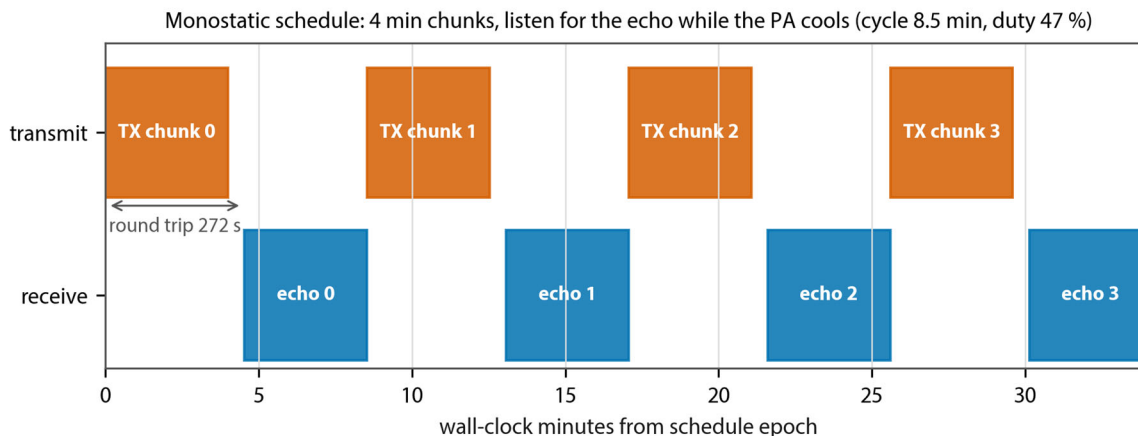
One message is one BCH codeword: 106 payload bits (90 message bits plus a 16-bit CRC), encoded to 127 bits, carried in 11 symbols of 12 bits each (132 bit positions, the last 5 zero). Each symbol is one tone held for 473 frames of 1/2.87 s. The numbers:

Quantity	Value	Derivation
Frame (FFT block)	0.34843 s	1 / 2.87 Hz
Symbol	164.808 s	473 frames
Message (11 symbols)	1812.9 s = 30.2 min	11 × 164.808 s
Payload rate	0.058 bit/s	106 / 1812.9
Message text	11 ASCII characters (90 bits, 8-bit packed)	

The frame is the unit everything is scheduled in. Frames are numbered from the schedule epoch; frame k carries symbol $\text{floor}(k / 473) \bmod 11$ of message repetition $\text{floor}(k / 5203)$. A transmitter sends whichever frames its schedule says are on; a receiver files every frame it receives into the accumulator of the symbol that frame belongs to. Because detection is non-coherent, frames of one symbol need not be contiguous in time, which is what makes chunking possible.

4.2 Monostatic chunking

DSES listens to its own echo, so the transmitter must be silent when the echo arrives. With round-trip time RTT , a chunk of length T_{on} transmitted from $t = 0$ returns during $[\text{RTT}, \text{RTT} + T_{\text{on}}]$. The next chunk may start at $\text{RTT} + T_{\text{on}}$. The DSES amplifiers add a duty limit of about 5 minutes on and 4 minutes off. The constraints are therefore $T_{\text{on}} \leq \min(\text{RTT}, 300 \text{ s})$ and $T_{\text{off}} \geq \max(\text{RTT}, 240 \text{ s})$, and the cycle is $T_{\text{on}} + T_{\text{off}}$.



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Figure 2 — Monostatic schedule with 4-minute chunks. Each chunk's echo returns 272 s after it was sent and is received while the amplifier cools; the next chunk follows the echo.

Shorter chunks are kinder to the transmitter but cost wall-clock time, because every chunk pays the full round trip in silence:

Chunk T _{on}	Cycle	Duty	Chunks per message	Wall-clock per message
1 min	5.5 min	18 %	31	167 min
2 min	6.5 min	31 %	16	100 min
3 min	7.5 min	40 %	11	78 min
4 min	8.5 min	47 %	8	64 min
4.5 min (= RTT)	9.0 min	50 %	7	59 min

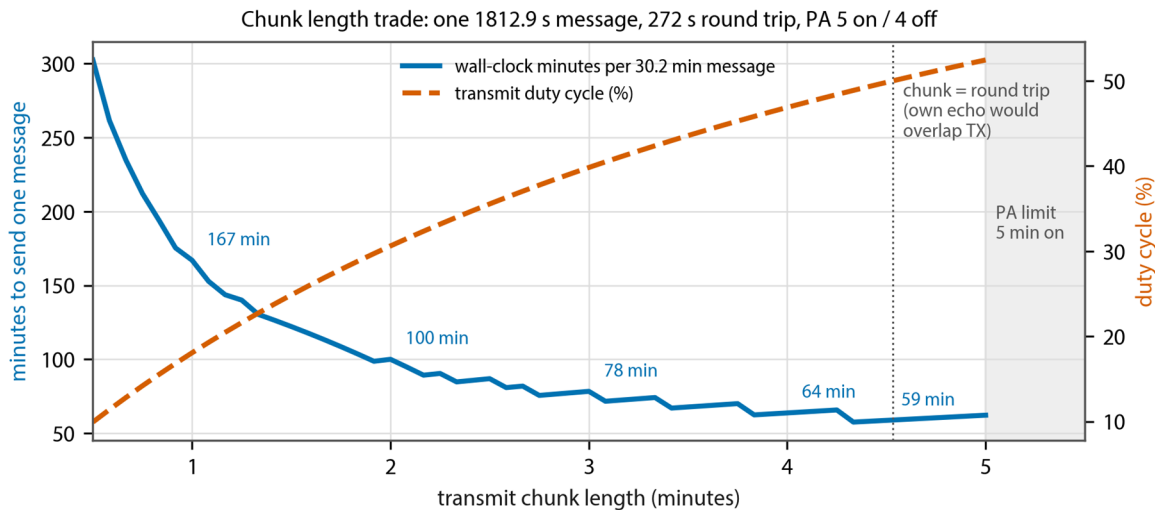


Figure 3 — Wall-clock time to send one 30.2-minute message and transmit duty cycle as a function of chunk length, for a 272 s round trip and the 5-on/4-off amplifier limit.

The working value is **4 minutes**: it keeps a 30-second guard below the round trip so a chunk's own echo never overlaps the next transmission, stays a minute under the amplifier's 5-minute limit, and delivers a message in 64 minutes. On conjunction day the 5.75-hour window above 20° elevation holds five message passes, which is what repeat-and-combine needs. Chunk boundaries fall on frame boundaries; a symbol may straddle a gap.

Symbol boundaries and chunk boundaries are independent. The alternative of making each chunk exactly one symbol (a 165 s chunk, 2.7 min) is a valid setting of the same parameter and is listed for completeness; it buys nothing the receiver needs and costs 20 percent in wall-clock time relative to 4-minute chunks.

4.3 Time and frequency reference

Pete's design assumes stations sharing a hydrogen maser over White Rabbit. The DSES receiver is non-coherent, so it needs two things only: to know which frame is which, and to have the tone land within a fraction of a 2.87 Hz bin for the whole symbol.

- **Time.** A GPS-disciplined 10 MHz and 1 PPS go into the B210's REF IN and PPS IN. The modem sets the USRP time to UTC on a PPS edge, so every transmitted sample and every received sample carries a UTC timestamp good to well under a millisecond. The schedule epoch is a UTC instant; frame boundaries are epoch + $k \times 0.34843$ s. The receiver maps arrival time to frame number through the ephemeris round-trip time. The tolerance is a fraction of a frame (tens of milliseconds); GPS provides microseconds.
- **Frequency.** The same GPSDO disciplines the B210's master clock. A B210 on its internal TCXO is ± 2 ppm, or ± 4.6 kHz at 2304 MHz with drift of tens of hertz per session; that alone would defeat a 2.87 Hz receiver. Locked to a GPSDO the reference is $1e-11$ or better, 0.02 Hz at 2304 MHz.
- **Ephemeris.** Round-trip time and Doppler come from a JPL Horizons table (topocentric, one row per second, fetched before the session and stored with the schedule), with astropy plus a local DE440s file as the offline fallback. The two are cross-checked before every session.

4.4 Doppler

The two-way Doppler at 2304 MHz on conjunction day sweeps from +9.5 kHz to -4.5 kHz across the pass and its rate peaks at -0.48 Hz/s (Figure 1). Across one 165-second symbol that is 79 Hz, 14 tone spacings; across one 0.35-second frame it is 0.17 Hz, six percent of a bin. So the frame is short enough that a tone stays put within a frame, and the symbol is long enough that the tone must be steered.

The modem does both halves:

- **Transmit pre-compensation.** The transmit NCO is offset by $-f_D(t)$ for the designated receiver's geometry, so the echo arrives at the nominal comb. For monostatic operation that is the Haswell two-way Doppler at the echo's arrival time, $t + RTT$. For bistatic operation it is the uplink Doppler at Haswell plus the downlink Doppler at the partner (the schedule file names the receiver so both ends agree on who compensates what).
- **Receive residual tracking.** The receive NCO applies the same model from the receiver's side, and a slow tracker corrects the residual from the model's error and the reference's drift using the energy of the detected tone. Frames are then on the design grid.

The Doppler model is a first-class part of the schedule file so that an offline receiver at another station can apply exactly what the transmitter assumed.

4.5 Repeat-and-combine and the pilot

Because the message is repeated within a session, the receiver keeps accumulators per symbol across repetitions. Deciding on the sum over N repetitions is equivalent to N times the frames per symbol, worth $10 \log_{10} N$ dB of sensitivity: two passes buy 3 dB, five buy 7 dB. The ORI README suggests exactly this; here it is designed in from the start and the schedule file carries the repeat count.

An optional **pilot** reserves a known tone for the first 40 frames (14 s) of each chunk. It costs 6 percent of throughput and lets the receiver verify the timing and Doppler model live, at the start of each chunk, rather than discovering after 64 minutes that the epoch was off by a frame. The pilot is on for the EME test and for the first Venus sessions; it is a schedule flag, not a waveform change.

4.6 The EME proof test

Everything above is exercised against the Moon before Venus, at very low power. What changes is parameterized, not rewritten:

Parameter	Venus	Moon
Round-trip time	272 s	2.5 s
Chunk length (monostatic)	≤ 272 s, working 240 s	≤ 2.5 s (7 frames), alternating TX and RX
Two-way Doppler	± 10 kHz, rate ≤ 0.5 Hz/s	± 4 kHz at 1296 MHz, rate small
Spread	2.87 Hz assumed	Libration spread at 1296 MHz can exceed 5.74 Hz; pick a low-libration window or set R_{bw} from the predicted spread
Link	C/N0 -4 to $+2$ dB-Hz by band and power	B210 output alone (+10 dBm) gives +12 dB-Hz at 1296 or 1299.5 MHz and +16 dB-Hz at 2304 or 2400 MHz (T_{sys} 60 / 76 K, mean Moon distance); 0 dBm gives +2 and +6
Frequency	1299.5 MHz (2026)	1296 or 1299.5 MHz and 2304 or 2400 MHz: both bands, no amplifier needed

The B210 alone is the transmitter for EME testing. Its +10 dBm output through the feed lands 12 to 16 dB above the design point, so the test is run at the design point by turning the B210 output down, or by adding calibrated noise to the archive offline, with no amplifier in the chain, on either band, as soon as a feed and a transmit/receive switch at the feed exist (O12). This decouples validation of the modulation and protocol from the amplifier schedules entirely, which is why it can start before the 23 cm retune and long before the 13 cm package.

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The EME test proves the schedule and keying at the fastest cadence the design will ever use, the GPS timing, the Doppler module with a different target, the receiver, and repeat-and-combine, end to end. Its switching rate (2.5 s alternation) is a station question recorded in section 10.

5. Software design

5.1 Module layout

The modem is a Python package `eve/` in this project, pure NumPy/SciPy in the core with GNU Radio only in the two blocks that touch the radio.

Module	Responsibility
params.py	EveParams: <code>R_bw</code> , <code>N_fft</code> , <code>M</code> , <code>N_frames</code> , <code>N_sym</code> , BCH <code>n/k</code> , IF offset, radio decimation, with the Python-Implementation values as defaults and the MATLAB set selectable for reproducing Pete's curve. Derived quantities (modem rate, frame, symbol, message time).
bch.py	BCH(127,106), $t = 3$, narrow-sense systematic, generator polynomial octal 11554743 over $GF(2^7)$ with primitive x^7+x^3+1 . Encoder and Berlekamp-Massey / Chien decoder. Verified against <code>galois</code> and against the Lin & Costello table.
message.py	Text $\leftrightarrow$ 90 bits (8-bit ASCII, zero-padded); CRC-16-CCITT (0x1021, init 0xFFFF); payload assembly; verification on decode.
modem.py	Symbol packing (MSB first, 132 positions), tone map, the streaming synthesizer (phase-continuous, chunk-gated, Doppler-offset NCO at any sample rate), and the receiver core: frame FFT bank, per-symbol magnitude accumulators, decisions, unpacking. Faithful to the ORI conventions; no Doppler or sync inside.
channel.py	Pete's <code>channel.m</code> (AWGN + random phase + Rayleigh, -1.05 dB) plus a streaming extension with Doppler ramp, timing offset, and gaps.
montecarlo.py	Success-rate vs C/N_0 and vs frames-per-symbol curves; the Rayleigh version of ORI's link check.
doppler.py	Two-way topocentric Doppler and round-trip time vs UTC for target venus or moon, from a Horizons table (primary) or <code>astropy</code> + DE440s (fallback); visibility window.

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schedule.py	The block-slot timeline: epoch, chunks, receive windows, frame-to-symbol map, repeat count, pilot flag; JSON in and out (section 8.1).
sync.py	Pilot detection, frame-grid search ($\pm N$ frames), residual frequency tracker, repeat-and-combine. Everything the reference design does not have, kept out of modem.py.
sigmf_io.py	Export in ORI's SigMF form (<code>ori:design</code> block) plus a <code>dses:</code> block; import of ORI files for playback.
gr_blocks.py	EveToneSource (schedule-driven NCO source at the radio rate) and EveRxSink (decimate to the modem rate, archive IQ, feed the live accumulators).
radio.py	B210 setup: external reference and PPS, <code>ref_locked</code> readback, UTC time set on PPS, IF-offset tune with readback verification, rate readback, GPIO keying line. Factored from the Workbench (section 9).
station.py	Key-down and cooldown interlocks, PA duty enforcement, abort, session log.
tools/	<code>eve_bench.py</code> (loopback soak, underrun and gap counters, offline noise injection), <code>eve_session.py</code> (run a schedule), <code>eve_decode.py</code> (offline decode of an archive).

5.2 Transmit path

EveToneSource produces samples at the radio rate directly: for sample n at time t_n it outputs $A \times \exp(j \times 2\pi \times (f_{IF} + d(k) \times \Delta f - f_D(t_n)) \times t_n + \phi)$, with $d(k)$ the symbol of frame k , and ϕ carried across symbol hops so phase is continuous. Outside an on-window it outputs zeros and the keying line is released. The block never touches a file; the schedule is its only input. Buffering follows the Workbench's proven practice for the B210 transmit edge (deep minimum output buffer, real-time mode during a chunk) so a 4-minute chunk runs without underruns; underruns are counted and logged.

The B210 TX/RX A port drives the amplifier chain; RX2 A takes the LNA. The two ports are always so assigned; the modem refuses to start if the receive port is on TX/RX.

5.3 Receive path

EveRxSink takes the B210 stream (radio rate = $32 \times$ modem rate, nominal 1.5047 MS/s), mixes down by the IF offset and comb centre, applies the receive Doppler NCO, decimates by 32 to the modem rate 47,022.08

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S/s, and writes the result to the session archive as complex64 with a JSON sidecar (section 8.2). At 376 kB/s a full 6-hour window is 8 GB; the archive is always written so any session can be re-decoded offline with a different schedule, Doppler model, or receiver. In parallel the live receiver runs the 16,384-point frame FFT bank, files frame magnitudes into the symbol accumulators, and shows the running symbol decisions and the tone-strip display.

The decoder is deliberately offline-first: the live view is for the operator; the decision of record is the offline decode of the archive after the session.

5.4 Validation plan

Stage	What	Gate
0	Reference vectors: ORI generator output for a fixed message; BCH round trip against galois	Bit-exact symbols and codeword
1	Modulator: DSES streaming synthesizer vs ORI file for the same message, same rate	Tones and phase match to float tolerance
2	Receiver in simulation: Pete's channel (Rayleigh) and ORI's AWGN model; frames-per-symbol sweep	Reproduces both curves; Rayleigh margin stated
3	Streaming realism: Doppler ramp, chunk gaps, wrong epoch; real CAMRAS Venus echoes (March 2025, public) through the FFT bank	Decodes at the design point with a 0.5 Hz/s ramp; pilot recovers a wrong epoch
4	B210 loopback on the bench: 4-minute chunk soak, underruns, reference lock, rate readback, transmit frequency vs the lab GPS reference on the E4438C / 53230A	Zero underruns, frequency within 0.1 Hz
5	EME at very low power	End-to-end decode at C/N0 near 0 dB with the station's own keying
6	Venus, sessions from mid October	

6. ICD part A — the air interface

This section is the interoperability specification. It follows the ORI Python implementation exactly where that implementation is definite, and it resolves, by definition, two places where it is not. Items marked TBC are awaiting confirmation from Pete Wyckoff and Michelle Thompson (asked 2026-09-09).

6.1 Waveform parameters

Parameter	Symbol	Value	Source and notes
Modulation		4096-ary orthogonal FSK, non-coherent	Pete Wyckoff, Spiral #2
FFT bin width = frame rate	R_bw	2.87 Hz	ORI Python; equals the Doppler-spread forecast at 2304 MHz, applied by ORI on every band. TBC: the MATLAB used 2.67 Hz; see O11 for 23 cm
Tone spacing	Δf	5.74 Hz (= 2 R_bw)	One guard bin between tones
Alphabet	M	4096 tones, 12 bits per symbol	
Frames per symbol	N_frames	473	ORI Python; its README notes Pete's slide shows about 440. TBC
Symbol duration	T_sym	164.808 s (= 473 / 2.87)	Defined as an integer number of frames. ORI Python uses 164.794 s (472.96 frames); the 14 ms difference is resolved here in favour of whole frames
Symbols per message	N_sym	11	
Message duration	T_msg	1812.9 s	
Occupied bandwidth		23.5 kHz (4096 × 5.74 Hz)	
FEC		BCH(127,106), t = 3, narrow-sense, systematic (message bits first)	Generator octal 11554743; identical in MATLAB and Python
Payload		90 message bits + CRC-	ORI Python

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		16-CCITT (0x1021, init 0xFFFF, no reflection) over the 90 bits, MSB first	
Message text		11 ASCII characters, 8 bits each, MSB first, zero-padded to 90 bits	90 bits = 11.25 characters; the 12th is truncated
Bit-to-symbol map		127 coded bits zero-padded to 132; symbol $m = \text{bits } 12m \dots 12m+11$, first bit most significant	ORI Python. The MATLAB used LSB first
Amplitude	A	constant, 0.8 of full scale in ORI files	Constant envelope; Class-C amplifier compatible
Phase		continuous through a symbol; at symbol hops DSES keeps phase continuous (ORI files restart from a global index). Receiver-invisible	

6.2 Tone map and spectral placement

Tone d ($0 \dots 4095$) is at baseband frequency $f_d = d \times \Delta f$ above the comb origin, one-sided, 0 to 23,505 Hz. The comb origin sits at an IF offset above the dial frequency so that tone 0 is not on the zero-IF spike:

$$f_{\text{RF}}(d) = f_{\text{dial}} + f_{\text{IF}} + d \times 5.74 \text{ Hz, with } f_{\text{IF}} = 25,000 \text{ Hz.}$$

Published operating parameters are therefore two numbers: f_{dial} (the frequency the transmitter is tuned to) and f_{IF} (25 kHz unless a schedule says otherwise). The comb occupies $f_{\text{dial}} + 25.0 \text{ kHz}$ to $f_{\text{dial}} + 48.5 \text{ kHz}$. A receiver tunes to f_{dial} , mixes down by f_{IF} plus the applicable Doppler, and sees tone d at $d \times 5.74 \text{ Hz}$. This is the ORI convention (their README: “tune the B210 25 kHz low”). The MATLAB placed the comb symmetrically about DC; that convention is not used on the air.

6.3 Receiver reference implementation

- Frame length $1 / R_{\text{bw}} = 0.34843 \text{ s}$. The modem sample rate is defined as $N_{\text{fft}} \times R_{\text{bw}}$ with $N_{\text{fft}} = 16,384$, giving 47,022.08 S/s, so a frame is exactly 16,384 samples and tone d is exactly FFT bin $2d$. Any radio rate is bridged to this by resampling (DSES: $32 \times$ modem rate at the B210, integer decimation).
- Per frame: FFT of the frame, magnitude at the 4096 candidate bins.

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- Per symbol: sum of frame magnitudes over the frames scheduled for that symbol (linear magnitude combining, as in Pete's `runTest.m`; power combining is an accepted alternative with equivalent performance and is what the ORI AWGN check assumes).
- Decision: argmax over the 4096 tones. Symbol 10 (the last) carries only 7 data bits; restricting its argmax to the 128 tones with the 5 padding bits zero is a documented option (off = faithful to ORI).
- Unpack 11 symbols MSB first to 132 bits, take the first 127, BCH-decode (up to 3 bit errors), split 90 + 16, check the CRC.

6.4 Frame and schedule timing

- Frames are numbered $k = 0, 1, 2 \dots$ from the schedule epoch t_0 (UTC); frame k occupies transmit time $[t_0 + k \times T_{\text{frame}}, t_0 + (k+1) \times T_{\text{frame}})$.
- Symbol index $m(k) = \text{floor}(k / N_{\text{frames}}) \bmod N_{\text{sym}}$; repetition $r(k) = \text{floor}(k / (N_{\text{frames}} \times N_{\text{sym}}))$.
- A transmitter emits frame k only if k lies in one of the schedule's on-windows.
- A receiver expects frame k at $t_0 + k \times T_{\text{frame}} + \text{RTT}(t)$, where RTT is evaluated at the transmit time from the schedule's ephemeris, and files it into accumulator $(r, m(k))$.
- Optional pilot: the first N_{pilot} frames (default 40) of every on-window carry tone d_{pilot} (default 2048) instead of the message symbol; the receiver excludes them from the symbol accumulators.

6.5 Doppler convention

The schedule names the receiver. The transmitter pre-compensates so that the tone arrives at the receiver at $f_{\text{RF}}(d)$ as defined in section 6.2: it transmits at $f_{\text{RF}}(d) - f_{\text{D}}(t)$, where f_{D} is the two-way (monostatic) or uplink-plus-downlink (bistatic) Doppler for the named receiver at the echo's arrival time. A receiver with the schedule file applies no bulk correction, only residual tracking. A receiver without the transmitter's schedule applies its own ephemeris for the full two-way Doppler.

7. ICD part B — the station

These are the interfaces between the modem software and the Haswell station. Values in square brackets are the modem's assumptions; the station team owns the actual numbers and the entries marked TBD need them.

7.1 Radio

Interface	Specification
Radio	Ettus USRP B210, one unit, owned exclusively by the modem process during a session. Tunes 70 MHz to 6 GHz; the dial frequency is a schedule parameter (1296, 1299.5, 2304, 2400 MHz, and others)
Transmit port	TX/RX A. Venus: to the amplifier drive chain; B210

	output [+10 dBm maximum], the modem runs $A = 0.8$ (−2 dB) at full TX gain unless the drive requirement says otherwise. EME tests: directly to the feed through the transmit/receive switch, no amplifier; output set by TX gain to the wanted C/N0 (section 4.6). TBD: required drive level at each PA input and any pad
Receive port	RX2 A, from the LNA. RX gain set for the receiver noise to sit 10–15 dB above the B210 floor (verified live with the tone-strip display)
Sample rate	1,504,706.56 S/s nominal ($32 \times$ modem rate); actual UHD rate read back and the residual absorbed by the frequency tracker
Tuning	f_{dial} with the verified LO-offset method inherited from the Workbench; the LO is parked off the comb so its leakage never lands on a tone
Reference	REF IN: 10 MHz from the station GPSDO, [+3 to +15 dBm, 50 Ω]. PPS IN: 1 PPS, 3.3 V logic. The modem refuses to start a session unless <code>ref_locked</code> reads true. TBD: which GPSDO is at Plishner and its outputs
Time	USRP time set to UTC at a PPS edge from the host's NTP/GPS time; verified against a second PPS before the session

7.2 Keying and sequencing

Interface	Specification
Keying output	One logic line from the B210 GPIO header (J504, 3.3 V, [FP0 bank, line 0]) through an isolated driver, or a USB relay if the station prefers. Asserted for the whole on-window
Sequencer timing	Key asserted [$T_{\text{lead}} = 200$ ms] before the first non-zero sample; RF stopped [$T_{\text{lag}} = 100$ ms] before the key is released. TBD: the station sequencer's actual lead and lag requirements and whether it protects the LNA
Duty limits	The modem enforces $T_{\text{on}} \leq [300 \text{ s}]$ and $T_{\text{off}} \geq [240 \text{ s}]$ per chunk regardless of the schedule; a schedule violating them is refused. TBD: the amplifiers' true thermal limits and whether 4-minute chunks are acceptable

Abort	Operator abort or any fault (reference unlock, underrun burst, USB error) releases the key immediately and logs the frame number
EME cadence	Monostatic Moon operation alternates transmit and receive every ≤ 2.5 s. TBD: whether the station's transmit/receive switching can follow that; if not, EME runs bistatic with a second receive antenna, or DSES transmits only and a partner receives

7.3 Feed and pointing

Interface	Specification
Feed	23 cm feed (1296 / 1299.5 MHz) for 2026; 13 cm feed (2304 / 2400 MHz) when that package exists. A feed change is a station operation; the modem does not control feeds
Pointing	The modem does not steer the dish. It computes and displays Venus azimuth and elevation from the ephemeris for the operator. The measured 0.15° boresight offset applies at 2304 MHz, where the beam is 0.14° wide; the pointing corrections memo of 2026-09-07 covers it

8. ICD part C — data and files

8.1 Schedule file

One JSON document per session, produced by `schedule.py` and shared with any partner station. It is the complete contract; a receiver needs nothing else.

```
{
  "schema": "dses-eve-schedule/1",
  "session_id": "DSES-EVE-20261024-A",
  "target": "venus",                      # venus | moon
  "epoch_utc": "2026-10-24T15:30:00Z",    # frame 0 transmit start
  "transmitter": {"site": "DSES Haswell", "lat": 38.380833, "lon": -103.156111,
    "alt_m": 1311},
  "receiver": {"site": "DSES Haswell", "mode": "monostatic"},
  "rf": {"f_dial_hz": 2304000000, "f_if_hz": 25000},
}
```

```

    "waveform": {"r_bw_hz": 2.87, "n_fft": 16384, "m": 4096, "n_frames": 473,
    "n_sym": 11,

        "bch": [127, 106], "crc": "CRC-16-CCITT", "bit_order": "msb_first",

        "amplitude": 0.8, "phase": "continuous"},

    "message": {"text": "DSES K0GD V", "payload_bits": "...106...", "codeword_bits":
    "...127..."},

        "symbols": [1269, 585, 516, 1366, 1166, 68, 1106, 112, 358, 2656,
    3072]},

    "repeat_count": 5,

    "pilot": {"enabled": true, "n_frames": 40, "tone": 2048},

    "chunks": [ {"index": 0, "frame_first": 0, "frame_last": 688,

        "tx_start_utc": "...", "tx_stop_utc": "...",

        "rx_start_utc": "...", "rx_stop_utc": "..."}, ... ],

    "doppler": {"model": "horizons", "table": "horizons_venus_haswell_20261024.csv",

        "convention": "tx_precompensated_for_receiver"},

    "limits": {"t_on_max_s": 300, "t_off_min_s": 240},

    "generated_by": "dses-eve 0.1", "generated_utc": "...
  }

```

The symbols listed are the transmitted symbols; a partner receiver may use them to score its decode directly.

8.2 Receive archive

<session_id>_<chunk>.eve.iq — complex64, little-endian, interleaved I/Q at the modem rate, one file per receive window, plus <session_id>_<chunk>.json with: start UTC from the USRP time, sample rate, f_{dial} , f_{IF} , the Doppler model that was applied on receive, frame number of the first sample, gain, reference-lock state, and overflow count. The archive is what the offline decoder reads and what is shared for independent decoding.

8.3 SigMF interchange

Export writes ORI's format: `core:datatype` `cf32_le`, `core:sample_rate`, the `ori:design` block with the same keys as ORI's generator, per-symbol annotations, and an additional `dses:schedule` block that

embeds the schedule file. Import accepts ORI's files as produced by `eve_tx_sigmf.py` and plays them through the DSES transmit chain unchanged, for cross-checks against the streaming synthesizer.

8.4 Session log and report

Every session writes a plain-text log (UTC, frame numbers, key events, underruns, overflows, reference state, aborts) and, after the offline decode, a one-page PDF report in the DSES house style: schedule summary, tone-strip image per symbol, decoded symbols against the transmitted ones, BCH corrections used, CRC result, and estimated C/N0 from the accumulators. The report is the deliverable of a session.

9. ICD part D — the Workbench boundary

The modem is a separate project. The interface to the DSES Radio Astronomy Workbench is a code-sharing boundary, not a runtime one:

- The Workbench's B210 classes (device discovery, `UhdB200Source`, the verified LO-offset tune path, the deep-buffer and real-time-mode helpers) are factored out of `dses_workbench.py` into an importable module that both projects use. The Workbench's behaviour does not change; this is a refactor the Workbench benefits from on its own.
- The Workbench's `FilterbankSink` pattern (a GNU Radio sink with a deep queue and a close method that reports gaps) is the template for `EverxSink`.
- The pulsar planner's site and visibility code is reused for the Venus and Moon windows.
- The Workbench keeps its rule that its own transmitter is locked at minimum gain. Real transmit drive exists only in the EVE modem, behind the interlocks of section 7.2.
- One B210 belongs to one process. During an EVE session the modem owns the radio; the Workbench is not running on it.

10. Decisions and open issues

10.1 Decisions taken in this revision

#	Decision	Rationale
D1	Stream the waveform; SigMF is export/import only	Chunking, live Doppler, no multi-GB files; nothing lost since schedule + symbols regenerate the IQ
D2	Monostatic baseline, 4-minute chunks	The only mode DSES controls; European receivers at 1299.5 MHz are the upside via the schedule file. 4 min stays under both the round trip and the PA limit; 64 min per message

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D3	ORI Python conventions are the air-interface spec; MATLAB set selectable for simulation only	ORI: the Python is the current implementation
D4	T_sym defined as 473 whole frames; modem rate defined as $16,384 \times 2.87$ Hz	Removes the two non-integer artefacts in the ORI numbers
D5	GPS time and GPSDO reference replace the maser	Non-coherent detection needs frame assignment and frequency, not phase
D6	Doppler pre-compensated on transmit for the named receiver, residual tracked on receive	79 Hz per symbol uncorrected
D7	Pilot frames on for EME and early Venus sessions	Cheap live check of timing and Doppler
D8	Receiver is offline-first; raw archive always written	Any session re-decodable with a better receiver
D9	Separate project; Workbench radio classes factored into a shared module	Different operational character; keeps the Workbench release train clean
D10	Phase-continuous symbol hops	Cleaner for the Class-C chain; invisible to a non-coherent receiver
D11	Frequency-agile modem: dial frequency per schedule; ORI waveform parameters unchanged on every band	23 cm in 2026, 13 cm later, EME on both; interoperability with the 1299.5 MHz stations
D12	EME tests use the bare B210 as the transmitter	+12 to +16 dB-Hz off the Moon with no amplifier; validation decoupled from amplifier schedules

10.2 Open issues

#	Issue	Owner	Needed by
O1	Confirm R_bw = 2.87 Hz and N_frames = 473 (vs 440 on the slide, 540 in MATLAB)	Pete Wyckoff / ORI	Before stage 2
O2	23 cm package: can it be retuned from 1296 to 1299.5 MHz; its power and Tsys (these set the	DSES station team	Before Venus

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	monostatic C/N0, -2 to -4 dB-Hz)		
O3	GPSDO at Plishner: model, 10 MHz level, PPS availability, cabling to the B210	DSES station team	Before EME test
O4	PA drive level required at the amplifier input; pad or preamp between B210 and PA	DSES station team	Before bench stage 4
O5	Keying interface: sequencer input type, lead/lag, LNA protection; whether 2.5 s alternation is possible for monostatic EME	DSES station team	Before EME test
O6	PA thermal limits: are 4-minute chunks with 4.5-minute cooldown acceptable for a 6-hour session?	DSES station team	Before Venus
O7	Solar noise at 6° separation: Tsys increase and its effect on the +1.7 dB-Hz margin	Link budget (DSES / ORI)	Before Venus
O8	Whether ORI wants the receiver contributed back to Python_Implementation	ORI	After EME
O9	Bistatic sessions with Effelsberg; if offered, the exact mutual window and who compensates Doppler	ORI / DSES	If offered
O10	EME libration spread at 1296 MHz on the test date; choose R_bw for the test	DSES	Before EME test
O11	R_bw at 23 cm: keep ORI's 2.87 Hz (interoperable, -2.8 dB) or a matched 1.5 Hz variant (incompatible with the European	ORI / DSES	Before Venus

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	receivers)? Recommendation: keep 2.87 Hz		
O12	Transmit/receive switch at the feed for direct-B210 EME tests: relay type, LNA protection, switching time for the 2.5 s alternation	DSES station team	Before EME test
O13	13 cm band: 2304 or 2400 MHz, and the feed for it	DSES station team	Next apparition

Appendix A — MATLAB simulation versus Python implementation

For readers of the ORI repository. The two describe different signals; the air interface in section 6 follows the Python.

Parameter	MATLAB (EveDemo.m, May 2026)	Python (eve_tx_sigmf.py, Aug 2026)
FFT bin / Doppler spread	2.67 Hz	2.87 Hz
Tone spacing	5.34 Hz	5.74 Hz
Frames per symbol	540	473 (README: slide shows about 440)
Symbol length	202.25 s	164.794 s (472.96 frames)
Message time	37.1 min	30.2 min
Bit-to-symbol order	LSB first ($2 \cdot (0:11)$)	MSB first
Comb placement	every other bin of an 8192-point FFT, symmetric about DC	one-sided, 0 to 23.5 kHz above the dial frequency + 25 kHz
Payload	106 random bits	90 message bits + CRC-16
Channel model	AWGN + Rayleigh ($\sigma = \sqrt{2/\pi}$, -1.05 dB), random phase per frame	none (analytic AWGN link check separate)
Receiver	simulated: magnitude sum over frames, argmax, BCH decode	none
BCH generator	octal 11554743 (MATLAB bchenc default)	same (galois default)

Appendix B — Timing arithmetic

- Frame: $1 / 2.87 \text{ Hz} = 0.348432 \text{ s}$; 16,384 samples at 47,022.08 S/s.
- Symbol: 473 frames = 164.808 s = 7,749,632 modem samples.
- Message: 11 symbols = 5,203 frames = 1,812.9 s.

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- 4-minute chunk: $240 \text{ s} = 688.8 \text{ frames}$; the schedule rounds chunks to whole frames ($688 \text{ frames} = 239.7 \text{ s}$), so a message needs $5,203 / 688 = 7.56 \rightarrow 8 \text{ chunks}$.
- Round trip 2026-10-24: $2 \times 40.8 \text{ million km} / c = 272 \text{ s}$; ephemeris-computed per session.
- Doppler across one symbol at the peak rate: $2304 \text{ MHz}, 0.48 \text{ Hz/s} \times 164.8 \text{ s} = 79 \text{ Hz} = 13.8 \text{ tone spacings}$; $1299.5 \text{ MHz}, 0.27 \text{ Hz/s} \times 164.8 \text{ s} = 45 \text{ Hz} = 7.8 \text{ spacings}$. Across one frame: 0.17 Hz (0.06 bin) and 0.09 Hz .
- EME link with the bare B210 (+10 dBm, 18.29 m dish, mean Moon distance, radar cross-section 6.5 percent of the disc): path loss 271.2 dB at 1296 MHz and 276.2 dB at 2304 MHz ; $C/N_0 = +12.2 \text{ dB-Hz}$ (Tsys 60 K) and $+16.2 \text{ dB-Hz}$ (Tsys 76 K).
- Radio rate: $32 \times 47,022.08 = 1,504,706.56 \text{ S/s}$; a 1 ppm rate error is 0.05 Hz at the top of the comb, absorbed by the residual tracker.

Document history

Revision	Date	Change
Rev A draft 1	2026-09-10	Initial design description and ICD
Rev A draft 2	2026-09-10	Band plan: Venus 2026 at 1299.5 MHz with the 23 cm package, 13 cm ($2304 / 2400 \text{ MHz}$) later, EME tests on both bands with the bare B210; link budget per band (section 2.2), section 2.4, D11 – D12, O11 – O13